Hospital Ward Occupancy Monitoring

Python Programming Case Study

1. Problem Statement

Hospitals need to monitor the number of patients occupying beds in each ward so that staff can identify high-demand periods and prevent wards from exceeding their available bed capacity. This case study uses Python lists and loops to display daily ward utilization, flag capacity exceedances, and identify the highest patient count recorded.

2. Objective

To develop a simple Python program that calculates daily bed-utilization percentages for General, ICU, and Emergency wards, reports when patient counts exceed bed capacity, and finds the peak occupancy along with the ward and day on which it occurs.

3. Assumptions

• The hospital monitors three wards: General, ICU, and Emergency.

• The bed capacities are 100, 20, and 30 respectively.

• Patient counts are recorded for three days for each ward.

• Each occupancy value represents the number of patients in that ward on that day.

• Utilization is calculated as (patients ÷ available beds) × 100.

• A patient count greater than the ward's bed capacity is reported as a capacity exceedance.

4. Algorithm

1. Start the program.

2. Create lists for ward names, bed capacities, and daily occupancy values.

3. Initialize the peak occupancy to zero.

4. Loop through each ward and display its name.

5. For each of the three days, read the patient count from the occupancy list.

6. Calculate utilization percentage and display the day and percentage.

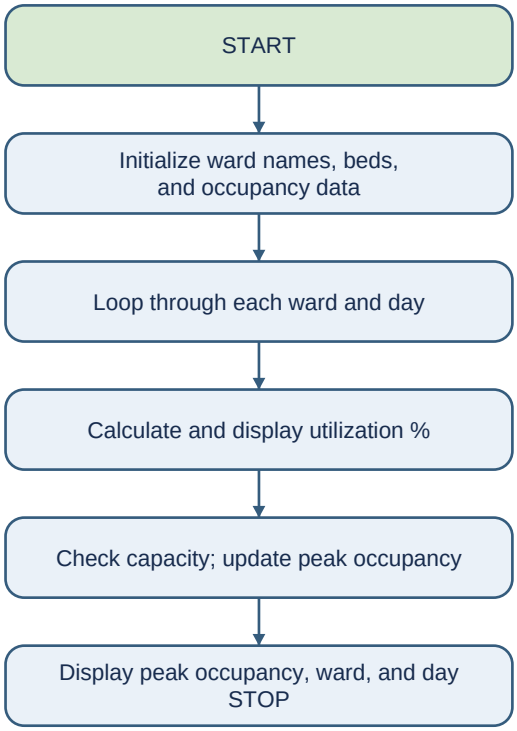
7. If patients exceed available beds, display a capacity-exceeded warning.

8. If the patient count is greater than the current peak, update the peak count, ward, and day.

9. After processing all wards, display the peak occupancy, ward, and day.

10. Stop.

5. Flowchart



Repeat the ward/day steps until all occupancy records are processed.

6. Pseudocode

```
BEGIN
    wards ← ["General", "ICU", "Emergency"]
    beds ← [100, 20, 30]
    occupancy ← [
        [60, 80, 90],
        [10, 15, 22],
        [28, 25, 28]
    ]

    peak ← 0

    FOR each ward index i
        PRINT ward name
        FOR each day index j
            patients ← occupancy[i][j]
            percentage ← (patients / beds[i]) × 100
            PRINT day and utilization percentage

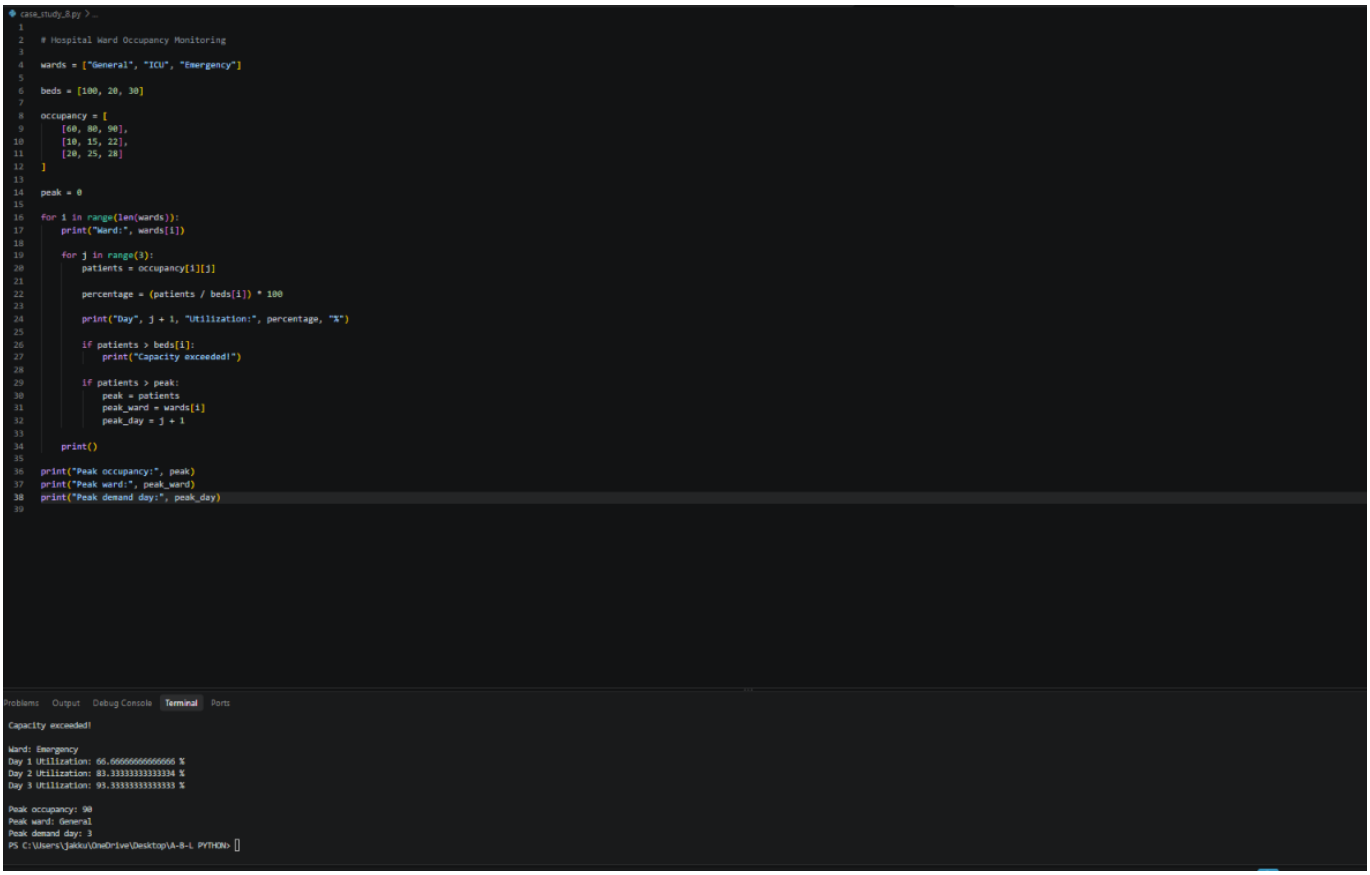
            IF patients > beds[i]
                PRINT "Capacity exceeded"

            IF patients > peak
                peak ← patients
                peak_ward ← wards[i]
                peak_day ← j + 1
        END FOR
    END FOR

    PRINT peak occupancy, peak ward, peak day
END
```

7. Source Code and Output Screenshot

The screenshot below contains the Python source code and the terminal output supplied for this case study.



8. Conclusion

The program demonstrates how Python can help monitor ward occupancy using lists, nested loops, conditional statements, and percentage calculations. It reports utilization for each day, warns when a ward exceeds its bed capacity, and identifies the highest patient count observed. In the provided data, the peak occupancy is 90 patients in the General ward on Day 3. The approach can be extended with live hospital data, graphical dashboards, and alerts for staff.